BOCCONI UNIVERSITY · COMPUTER VISION & IMAGE PROCESSING

Feedback-Augmented RT-DETR

The story of a silent mechanism, made measurable.

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1

Small objects are hard.

RT-DETR-R50 reports $\mathbf{AP_S = 34.7}$ on COCO val2017 — *thirty-three points* below $\mathbf{AP_L = 67.7}$ on the same network.

Why? Encoder memory is computed once. The decoder cannot revise it.

2

Let the decoder feed back.

After layer 1 makes preliminary predictions, refine the encoder memory with cross-attention:

$$\mathbf{m}' = \mathbf{m} + g \cdot \mathrm{Attn}(\mathbf{m}, \mathbf{t}^{(1)})$$

Layers 2..L then read the refined memory. $g \in (0,1)$ is a learnable scalar gate.

3

v1 trained — and contributed nothing.

Plain sigmoid gate, all four pyramid levels. Same-checkpoint ablation:

$$\Delta \mathbf{AP_S} = \mathbf{0.00}$$

The gate decayed to ≈ 0.12. The cross-attention output was multiplied by zero before reaching memory.

4

Two structural fixes. Zero new parameters.

(i) Gate floor. $g_{\text{eff}} = 0.1 + 0.9\,\sigma(\alpha)$ — *the optimizer can attenuate, never silence.*

(ii) Level mask: P2/P3 only. Refine stride-4 and stride-8 features (where small objects live); skip P4, P5.

Same-checkpoint inference ablation:

$$\Delta \mathbf{AP_S} = \mathbf{+0.99}$$

33.26 → 34.25, only inference behavior changed